QAOA for MaxCut

A scaling study on 3-regular graphs

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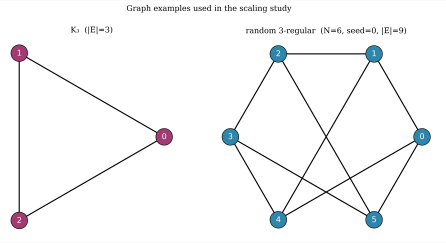
MaxCut — the classical problem

Given an unweighted, undirected graph $G = (V, E)$, find a partition $S \subseteq V$ that maximises the number of *crossing* edges:

$$C(S) = \left|\{(u, v) \in E : u \in S,\ v \notin S\}\right|,$$

$$C^* = \max_{S \subseteq V} C(S).$$



Two representative instances from this study.

Why it is a good benchmark

- NP-hard; no efficient classical exact algorithm known.
- Best classical approximation: Goemans–Williamson, ratio ≈ 0.878 (1995).
- NISQ-friendly: graph is sparse, Hamiltonian is diagonal, circuit depth scales gently.

From bits to spins — the Ising encoding

Binary $x_i \in \{0,1\}$, spin $s_i = 1 - 2x_i \in \{\pm 1\}$ (so $x_i{=}0 \Leftrightarrow i \in S$). Substituting $x_i = (1 - s_i)/2$ in the cut objective and promoting $s_i \to Z_i$:

$$\boxed{H_C \;=\; \frac{1}{2} \sum_{(i,j)\in E} \left(Z_i Z_j - \mathbb{1} \right)}$$

Every computational basis state is an eigenstate: $H_C \left|z\right\rangle = -C(z) \left|z\right\rangle$ — the ground state is the optimal cut.

K_3 spectrum (our toy problem)

eigenvalue	mult.	$C(z)$	states
-2	6	2	cuts
0	2	0	$\left\|000\right\rangle, \left\|111\right\rangle$

Uniform baseline (K_3): $\langle\psi_0| H_C |\psi_0\rangle = -\frac{3}{2}$, $r = 0.75$. Any non-trivial quantum algorithm must beat this.

Convention: H_C minimisation form (Qiskit-compatible); Farhi 2014 uses the opposite sign.

The QAOA ansatz

$$|\psi_p(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle \;=\; U_B(\beta_p)\,U_C(\gamma_p)\,\cdots\,U_B(\beta_1)\,U_C(\gamma_1)\;H^{\otimes n}\,|0\rangle^{\otimes n}\,.$$

Cost operator

$$U_C(\gamma) \;=\; \prod_{(i,j)\in E} e^{-i\frac{\gamma}{2}Z_iZ_j}$$

Per edge, $R_z(\theta) = e^{-i\frac{\theta}{2}\sigma_z}$:

$$e^{-i\frac{\gamma}{2}Z_iZ_j} = \mathrm{CX}_{i,j}\,R_z(\gamma)_j\,\mathrm{CX}_{i,j}$$

$\Rightarrow 2|E|$ CNOTs per layer.

Mixer operator

$$U_B(\beta) \;=\; \prod_{i=0}^{n-1} e^{-i\beta X_i} \;=\; \prod_i R_x(2\beta)_i$$

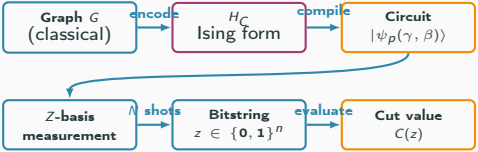
$R_x(\theta) = e^{-i\frac{\theta}{2}\sigma_x}$ gives the factor 2β.

$\Rightarrow n$ single-qubit gates, all parallel (depth 1).

Objective $F_p(\boldsymbol{\gamma},\boldsymbol{\beta}) = \langle\psi_p|\,H_C\,|\psi_p\rangle$ — classical optimiser minimises, quantum device evaluates.

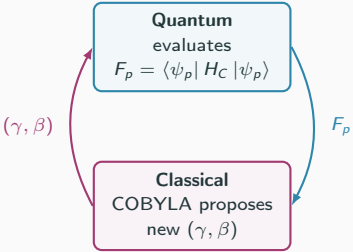
I/O pipeline and the variational loop

Data encoding → measurement



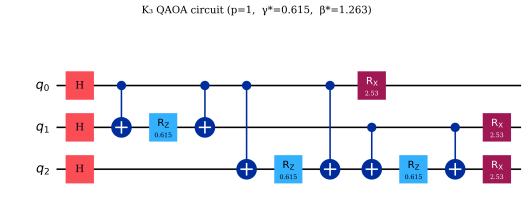
Input: graph → diagonal Hamiltonian $H_C = \frac{1}{2}\sum_{(i,j)\in E}(Z_i Z_j - I)$.
Output: best bitstring z^* sampled from $|\langle z|\psi_p\rangle|^2$; report $C(z^*)$.

Hybrid quantum–classical loop

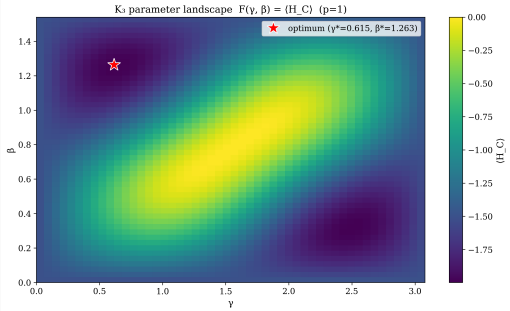


Classical side minimises F_p; quantum side is only called as a black-box expectation oracle. Parameters stay low-dim ($2p$ floats), so COBYLA converges in tens–hundreds of calls.

K_3 toy — circuit and parameter landscape



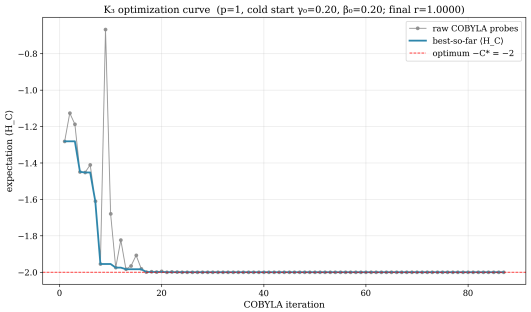
$p = 1$: $3\,H + 3\times(\mathrm{CX}\text{–}R_z\text{–}\mathrm{CX}) + 3\,R_x(2\beta)$.



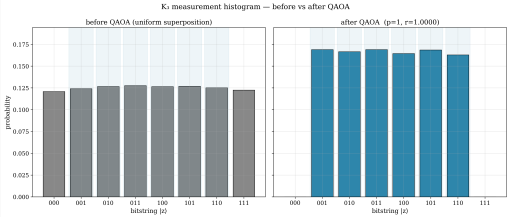
$F(\gamma,\beta)$ over $[0,\pi)\times[0,\pi/2)$. The two equivalent minima are related by the symmetry $F(\gamma,\beta) = F(\pi-\gamma,\pi/2-\beta)$.

Both minima reach $\langle H_C\rangle = -2 = -C^*$, i.e. $r = 1.0$ — $p = 1$ is already sufficient for K_3.

K_3 toy — convergence and measurement



COBYLA from a cold start $(\gamma_0, \beta_0) = (0.20, 0.20)$; grey = raw probes, blue = best-so-far.



20,000 shots; $|000\rangle$ and $|111\rangle$ suppressed to ≈ 0, the six optimal cuts lift from $1/8$ to $1/6$.

Final $r = 1.0000$ — the quantum state concentrates all probability on the optimal cuts.

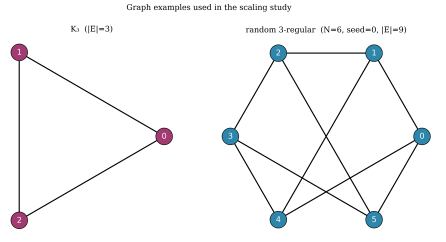
Scaling study — setup

Experiment grid

- $N \in \{3, 4, 6, 8, 10\}$ vertices
  - $N = 3$: K_3 (triangle, 2-regular)
  - $N \in \{4, 6, 8, 10\}$: random 3-regular
- Depth $p \in \{1, 2, 3\}$
- 5 seeds per (N, p)
- **Total: 75 experiments**

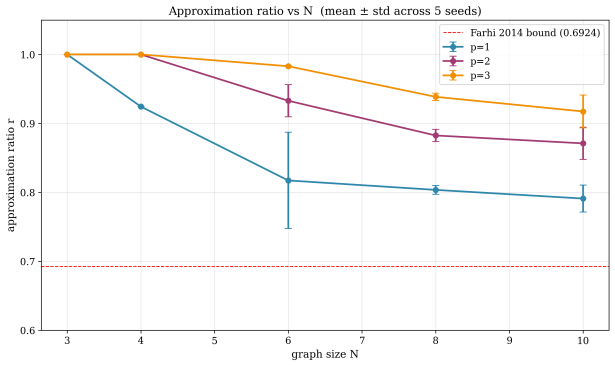
Optimisation

- $p = 1$: 15×15 grid warm start $\rightarrow$ COBYLA refine.
- $p > 1$: freeze prior layers, 10×10 mini-grid on new (γ_q, β_q) $\rightarrow$ joint COBYLA.



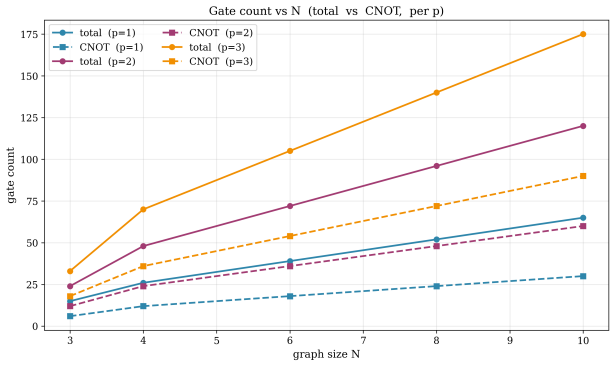
Metrics collected per run:
r (exact via Statevector), CNOT count, total gate count, circuit depth (post-transpile), COBYLA iterations, wall time.

Scaling results — approximation ratio vs N



- All 75 runs clear the Farhi 2014 bound $r \geq 0.6924$ ($p = 1$, 3-regular).
- Tightest margin: $N = 6$ seed 1, $r = 0.6925$.
- $N = 4$: all 3-regular graphs are K_4 — *zero* seed variance.
- $N = 6$: widest variance, diverse 3-regular topology with varying QAOA hardness.
- $N \geq 8$: parameter concentration (Brandão et al. 2018; Blekos et al. 2024) narrows the variance — seeds converge to similar optima.
- Degradation with N at fixed p: more edges, same 2-parameter-per-layer ansatz.

Scaling results — gate cost vs N



Per layer on a 3-regular graph ($|E| = 3N/2$):

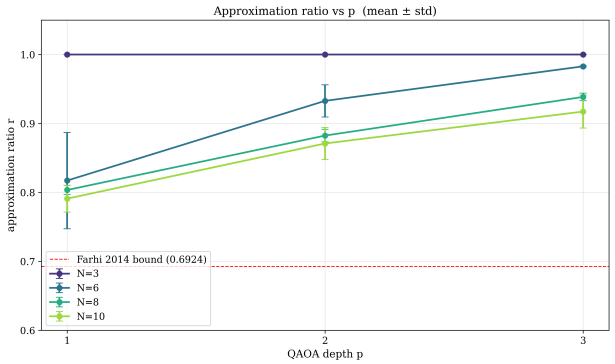
$$2|E|\ \mathrm{CX} \ + \ |E|\ R_z \ + \ N\ R_x$$

plus an initial $N\,H$. CNOTs scale linearly in N, yielding $3N\,p$ two-qubit gates at depth p.

For $N = 10,\ p = 3$: **90 CNOTs**, total depth 66 after transpile to $\{cx, rz, rx, h\}$ — comfortably within current NISQ coherence budgets.

Depth scales sub-linearly: parallelisable CNOT scheduling on sparse 3-regular graphs keeps depth below the raw CNOT count ($66 < 90$ at $N = 10, p = 3$).

Depth frontier — r vs p



Deeper ansätze monotonically close the approximation gap: monotonicity holds on **all 25** (N, seed) pairs, zero violations.

- $N = 3$ saturates at $p = 1$.
- $N = 10$, $p = 3$: $r = 0.917 \pm 0.024$.
- Beats Goemans–Williamson (0.878) already at $p = 2$ for $N \leq 6$.

Trade-off: each added layer costs another $2|E|$ CNOTs and $\propto$ COBYLA wall time.

Conclusions

- **Validated QAOA** on 75 MaxCut instances spanning $N \in \{3, 4, 6, 8, 10\}$, $p \in \{1, 2, 3\}$, 5 seeds each.
- **Theoretical lower bound respected:** every $p = 1$ run on 3-regular graphs meets $r \geq 0.6924$ (Farhi 2014).
- **Monotonicity** $r(p{=}1) \leq r(p{=}2) \leq r(p{=}3)$ holds on every (N, seed) pair.
- **Hardware-friendly scaling:** $3Np$ CNOTs on 3-regular graphs, linear in N.
- **Best result:** $N = 10,\ p = 3 \Rightarrow r = 0.917$, beating the classical Goemans–Williamson benchmark.
- The $p \to \infty$ optimality limit is theoretical; in the NISQ budget $p \in \{1, 2, 3\}$ already delivers $r \geq 0.87$ on all tested instances.

Take-away. Even at $p = 3$ QAOA is a credible candidate for near-term MaxCut on sparse graphs, with predictable gate and depth budgets.

References

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Individual project (no co-authors).
Source code, data, and figures: `github.com/ozgunyurutken/qaoa-project`